

Where hydrogen can win the dispatch

On the full annual cost of heat the heat pump is cheapest in 22 of 29 countries, so heat electrifies. Hydrogen enters only as a peaker, priced on short-run marginal cost against the carbon-taxed competitor, and is tested in three linked arenas. The count is the number of the 29 countries in which it would be dispatched.

1. Building winter peak

vs the carbon-taxed gas boiler
at the peak hour



Curr. Stated Net Zero H2 Push
rising policy ambition

electrify



2. District-heat stack

vs gas CHP on the marginal
heat unit



Curr. Stated Net Zero H2 Push
rising policy ambition

winter peak



3. Power sector

H2 turbine vs gas peaker,
€/MWh-e



Curr. Stated Net Zero H2 Push
rising policy ambition

Mechanism

The carbon price on the competitor, not cheap hydrogen, opens the niche, and the counts scale with ambition because carbon pricing does. At the power peak the H2 turbine (about 221) sits just below gas (about 239 €/MWh-e).

Caveat

Even where it wins, the peaker runs too few hours to recover its capital, about 15% under H2 Push. That is the classic missing-money problem, so the role is a dispatched one, not one a market would build unaided.